Safe System of Work
Recovery of Trapped Wagons

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Reviewed By: Name	Derick Bibby	Date of Review:	29.08.2026
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Overview

The following information provides guidance on the company's Safe Systems of Work for the Recovery of Trapped Wagons.

Wagons can become trapped in mud, which can present a difficult and dangerous job to get them moved. The task of recovery needs to be properly managed and controlled.

All employees have a responsibility for their safety and the safety of others by adhering to these safety systems.

AUTHORISED PERSONNEL ONLY TO PARTICIPATE IN THE RECOVERY ACTIVITIES

Hazards Identified

- Crushing
- Cuts and abrasions
- Adverse weather
- Vehicle movements
- Manual handling
- Slips & Trips
- Snap-back of chains, straps or shackles under load

Personal Protective Equipment (PPE) Requirements









Operations

Upon arrival at site:

Report to the site office and sign in.

The recovery operative will take charge of the task and the wagon driver will cooperate fully.

Make sure that the site manager is aware that the wagon is to be recovered and all persons not involved to be clear of the area.

Initial Assessment

Is the wagon recoverable with the equipment available?
If not, contact third party recovery company.
If recovery operator concludes that the wagon is recoverable, they will start the process to move it.

Must consider:

Condition of wagon – Is it road worthy? Will it need to be towed back to the yard? Is it clean enough to be driven/towed on the public highway?

Weather – Is heavy rain or snow going to impede the recovery process?

Ground conditions – Is the wagon stuck in deep mud? How much mud is there surrounding the wagon? Once moved, can it move on its own, or will it need towing to another area?

Level ground – Is the wagon on a level surface? Is the wagon sunk in at an angle? Is there level ground from which to make a safe recovery?

Overhead hazards – Are there any overhead cables, GS6 goalposts or bridges that will prevent recovery in a certain direction?

Passing traffic – Are there moving vehicles close by? Is there a pedestrian walkway nearby? Is the wagon currently adjacent to a public highway? Is there any mobile plant passing close by?

Slips/Trips – Is the recovery area accessible by foot? Are the any serious concerns regarding access by foot?

Equipment – Size of excavator? Capability of chains/straps? Condition of equipment? Is all equipment fit for purpose?

Equipment:

360 Excavator

- CPCS Operative
- Suitable for job (weight capacity)
- Safe condition (daily checks done, no issues flagged)
- In-date certification

Chains/Straps

- Correct weight rating (metal tag, strap label)
- Pre-use visual checks (no cracks, stretches, warps etc)
- Good condition

Recovery:

If recovery with excavator and chains/straps is suitable and safe, the recovery operative can begin the process of freeing the wagon.

Initially, an attempt may be made to nudge the wagon out, using the designated bar across the back of the wagon. If this does not work, use the following steps:

- Clear immediate area of pedestrians, vehicles, and machines that are not part of recovery process.
- Excavator operative to manoeuvre machine into place, as directed by recovery operative.
- Both wagon and excavator to be switched off and isolated, with hand brakes applied.
- Shackle to be fixed to excavator, in correct position.
- Chain/Strap to be fixed to towing eye on wagon
- Chain/Strap to be fixed to shackle on excavator
- Recovery operative to move back to safe area, ensuring line of sight with all parties involved.
- Recovery operative to then provide step-by-step instructions to excavator operative and wagon driver, dependant on the individual situation and how they wish to proceed.
- Wagon will be freed and then moved to a safe area.

Once the wagon has been moved to a safe area, an assessment must be made of any damage caused and roadworthiness.

No machines or wagons shall be allowed on the public highway in a dangerous or unsafe condition. If there is any damage that gives cause for concern, a recovery will be undertaken.

Snap-Back and Exclusion Zone:

A chain, strap, shackle or towing eye that fails under load can whip back with enough force to kill or seriously injure anyone in its path. The following controls apply to every recovery:

- Only use equipment rated and marked for towing or recovery (towing chains, recovery straps, bow shackles) with a capacity suitable for the gross weight of the wagon.
- Lifting chains, slings and other lifting accessories must never be used for towing or recovery.
- Only attach to the manufacturer's towing eyes or rated recovery points – never to axles, springs, bumpers or tow balls.
- Set an exclusion zone of at least 1.5 times the length of the chain/strap on all sides of the line of pull. Nobody may enter the zone or stand in line with the chain/strap while it is under tension.
- The wagon driver stays in the cab with the door closed; the recovery operative directs from outside the zone.
- Fit a recovery damper or blanket over the centre of the chain/strap to absorb energy if it fails.
- Take up the slack slowly and pull steadily – no snatching or jerking.
- Stop if equipment shows damage or stretching, or an anchor point deforms. Quarantine damaged or overloaded equipment.